Introduction to Golang

Concepts of Programming Languages
4 October 2018

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Gopher



Why Go?

- Go is the major language behind the Cloud Native Stack

www.cncf.io/ (https://www.cncf.io/)



- The most important components are written in Go: Docker, Kubernetes, etcd, Prometheus, Grafana, ...
- Go is Open Source and maintained by Google
- Go is a distributed, parallel language designed for systems programming at Google to solve the problems of C++ code.

Hello World

```
// Copyright 2018-2026 Johannes Weigend
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package main

import "fmt"

func main() {
    fmt.Printf("Hello %s", "Programming with Go \xE2\x98\xAF\n") // \xE2\x98\xAF -> ☯
    fmt.Printf("Hello %s", "Programming with Go ☯\n")
}
```

Run

Palindrome

```
// IsPalindrome implementation. Does only work for 1-Byte UTF-8 chars (ASCII).
func IsPalindrome(word string) bool {
    for pos := 0; pos < len(word)/2; pos++ {
        if word[pos] != word[len(word)-pos-1] {
            return false
        }
    }
    return true
}
```

```
// palindrome_test.go
func TestPalindrome(t *testing.T) {
    if !IsPalindrome("") {
        t.Error("isPalindrome('' should be true. But is false.")
    }
    if !IsPalindrome("o") {
        t.Error("isPalindrome('o' should be true. But is false.")
    }
    if !IsPalindrome("oto") {
        t.Error("isPalindrome('oto' should be true. But is false.")
    }
    if IsPalindrome("ottos") {
        t.Error("isPalindrome('ottos' should be false. But is true.")
    }
```

Introduction to Golang

Rob Pike @ Google 2009 (60 Min)

www.youtube.com/watch?v=rKnDgT73v8s (https://www.youtube.com/watch?v=rKnDgT73v8s)

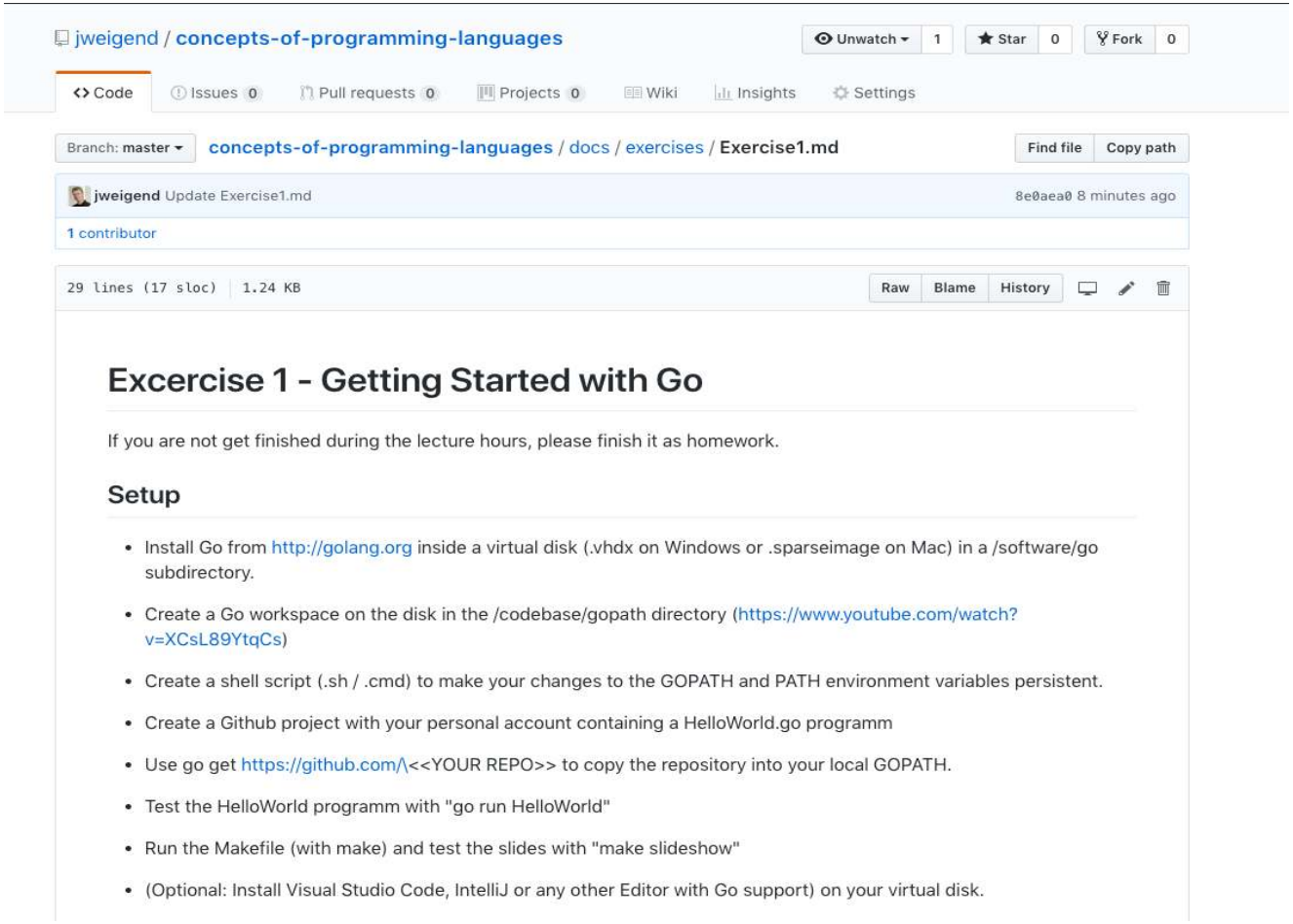


Some Questions

- What makes Go different to other Languages?
- What makes Go similar to other Languages?
- Discuss your personal opinions in a group of students!

Exercise 1

github.com/jweigend/concepts-of-programming-languages/blob/master/docs/exercises/Exercise1.md (https://github.com/jweigend/concepts-of-programming-languages/blob/master/docs/exercises/Exercise1.md)



See also

- github.com/jweigend/concepts-of-programming-languages (https://github.com/jweigend/concepts-of-programming-languages)
- golang.org/ (https://golang.org/)
- golang.org/doc/ (https://golang.org/doc/)

Thank you

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